+1 (412) 330 9634
ajreddy@cs.cmu.edu
anikethjr.github.io
linkedin.com/in/aniketh-reddy

Aniketh Janardhan Reddy

Education

Carnegie Mellon University - School of Computer Science, *Pittsburgh, PA*, Expected Dec 2019
Master of Science in Machine Learning, QPA: 4.08/4.33.
Coursework: Introduction to Machine Learning (PhD), Probability and Mathematical Statistics, Machine Learning for Large Datasets, Topics in Deep Learning, Probabilistic Graphical Models, Advanced Machine Learning: Theory and Methods

Birla Institute of Technology and Science (BITS), Pilani, *Hyderabad, India*, Jul 2018
Bachelor of Engineering (Honours) Computer Science, CGPA: 9.93/10.

Experience

Software Enginner Intern, Microsoft, *Sunnyvale, CA*. May 2019–Aug 2019
- Devised a new demand-based autoscaler for Application Gateways as a member of the Azure Networking group. This led to a $\sim 66\%$ reduction in the number of dropped requests compared to the previous system.

Research Assistant, Carnegie Mellon University, *Pittsburgh, PA*. Aug 2018–Present
- Working with Prof. Leila Wehbe on understanding the representation of language syntax in the human brain using functional magnetic resonance imaging (fMRI) and machine learning. After establishing that effort-based metrics widely used to study syntax processing are inadequate, we proposed subgraph embedding-based features which encode more meaningful syntactic information as alternatives and showed that they are more predictive of brain activity than the effort-based metrics.
- We are currently exploring the usage of BERT and ELMo for deriving more syntactic features.

Intern, Smart Data Analytics Research, Universität Bonn, *Bonn, Germany*. May 2017–Aug 2018
- Worked on several projects in the broad area of automated fact verification under the tutelage of Prof. Dr. Jens Lehmann.
- Developed a fact verification system called DeFactoNLP which could not only assess the veracity of a claim but also retrieve supporting evidences from Wikipedia. The system made use of named entity recognition (NER), TF-IDF vector comparisons and a Decomposable Attention-based natural language inference model. It achieved a 0.4277 evidence F1-score and a 51.36% label accuracy which were 134% and 5% above the baseline scores respectively.
- Built a state-of-the-art system which could ascertain the credibility of a website using handcrafted features derived from the text and visuals present on the website. Also, co-developed and benchmarked DeFacto, a triple verification framework.

Research Intern, Cognition Lab, Indian Institute of Science (IISc), *Bangalore, India*. Jan 2018–May 2018
- Used fMRI and machine learning to identify brain regions involved in attention modulation under the guidance of Prof. Sridharan Devarajan. This was accomplished by analyzing which regions' activations were most predictive of times at which a participant was required to pay covert attention. Many deep learning-based seq2seq models were tested in this context.

Academic Projects

Incorporating Visual and Textual Cues in Dialogue Generation for Comic Strips. Jan–May 2019
- Created two systems - one based on Visual Story Telling models and another based on Conditional Variational Autoencoders, which can be used to generate dialogues for comic strips given their images.

Object Detection, Removal and Inpainting using Deep Learning. Jan–May 2019
- Developed a system which makes use of a Mask RCNN (Regional Convolutional Neural Network) to detect an unwanted object and inpaints it using a contextual attention-based generative inpainting model.

Detecting Duplicate Questions using Siamese Networks and Decomposable Attention. Aug–Dec 2018
- Devised a system which determines if any two Quora questions are semantically equivalent using Decomposable Attention and a Long Short-Term Memory (LSTM)-based Siamese network. An accuracy of 83.98% was obtained using this system.

Named Entity Recognition (NER) for Telugu using LSTM-CRF. Aug–Dec 2017
- Built a system which used LSTMs and Conditional Random Fields (CRFs) for NER in Telugu texts with a 0.8513 F1 score.

Honors and Certifications

Gold Medal, for best academic record (highest CGPA), BITS Pilani, Hyderabad, India. Aug 2018

Best All-Rounder Award, in recognition of multidimensional achievements, Aug 2018
Prof. V.S. Rao Foundation and BITS Pilani, Hyderabad, India.

Technology Entrepreneurship Program, Indian School of Business (ISB), Hyderabad, India. Jun 2018

Merit Scholarship, BITS Pilani, Hyderabad, India. 2014–18

Working Internships in Science and Engineering Scholar, German Academic Exchange Service. 2017

Skills

Languages: Python, C/C++, Java, MATLAB, SQL **Libraries:** PyTorch, Tensorflow, NumPy, SciPy

Publications

- **Aniketh Janardhan Reddy**, Gil Rocha, and Diego Esteves. 2018. DeFactoNLP: Fact Verification using Entity Recognition, TFIDF Vector Comparison and Decomposable Attention. FEVER 2018 at EMNLP 2018. [Paper][Code]
- Diego Esteves, **Aniketh Janardhan Reddy**, Piyush Chawla, and Jens Lehmann. 2018. Belittling the Source: Trustworthiness Indicators to Obfuscate Fake News on the Web. FEVER 2018 at EMNLP 2018. [Paper][Code]
- **Aniketh Janardhan Reddy**, Monica Adusumilli, Sai Kiranmai Gorla, Lalita Bhanu Murthy Neti, and Aruna Malapati. 2018. Named Entity Recognition for Telugu using LSTM-CRF. WILDRE4 at LREC 2018. [Paper][Code]
- Diego Esteves, Anisa Rula, **Aniketh Janardhan Reddy**, and Jens Lehmann. 2018. Toward Veracity Assessment in RDF Knowledge Bases: An Exploratory Analysis. J. Data and Information Quality 9, 3, Article 16 (February 2018). [Paper]